RONIT MIA

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SUMMARY

Computer Engineering graduate with internship experience building Python computer-vision pipelines and a geospatial RAG application. Project experience spans backend services, retrieval systems, databases, authentication, and data pipelines.

TECHNICAL SKILLS

Languages: Python, Java, C++, SQL, JavaScript, TypeScript

Backend & Data: Spring Boot, REST APIs, Flask, Django, Node.js, PostgreSQL, MySQL, Redis, Qdrant, PySpark

Tools & Cloud: Git, Linux, Docker, Kubernetes, AWS, Terraform

Additional: Microservices, Authentication, RBAC/ABAC, RAG, Vector Search

EXPERIENCE

AI & Machine Learning Intern | AGT Intelligence Pvt. Ltd. May 2025 – Jul 2025

- Built Python computer-vision pipelines for geospatial infrastructure analysis across 20+ cities, processing 5TB+ of aerial imagery and reducing manual analysis time by 40%.
- Used Segment Anything Model (SAM) and DINOv2 to extract building footprints and spatial features; reported 95% precision in automated site selection.
- Built a spatial RAG system using Qdrant and semantic retrieval to support a geospatial chatbot answering location-specific queries.

PROJECTS

Contextual Memory Framework

- Built a multi-tenant knowledge-retrieval platform using Spring Boot microservices, PostgreSQL, Redis, Qdrant, Kubernetes, and S3.
- Implemented hybrid retrieval using BM25, dense vector search, Reciprocal Rank Fusion, cross-encoder reranking, and MMR.
- Added RBAC/ABAC tenant isolation and complexity-based LLM routing with Redis caching.

Real-Time Sports Analytics Platform

- Built a live sports platform for 30 teams with standings, box scores, play-by-play updates, 10-second polling, and deterministic game simulation.
- Implemented authentication with scrypt, HMAC-SHA256-signed HttpOnly cookies, route protection, and PostgreSQL-backed user profiles.

Data Engineering Medallion Platform

- Developed a PySpark Bronze–Silver–Gold pipeline processing 1.8M+ rows into seven curated datasets after selection for Citi's Data Engineering Coding Workshop.
- Added schema and foreign-key validation, deduplication, quarantine handling, structured logging, automated lineage, and 48 tests; deployed to AWS EKS using Terraform.

EDUCATION

COEP Technological University — B.Tech, Computer Engineering | Pune, India | 2022–2026

Relevant coursework: Data Structures & Algorithms, Database Systems, Operating Systems, Computer Networks